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Impact of ChatGPT on learning outcomes and performance of students in computer programming courses: a mixed-method approach

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Abstract
The rapid post-2022 integration of generative AI tools, such as ChatGPT, into Nigerian higher education necessitates an empirical evaluation of their impact on programming courses. This study examines the impact of ChatGPT on student learning outcomes and performance, addressing concerns related to critical thinking and academic integrity. We combined student and instructor surveys with course performance data using a mixed-methods approach within a mandatory undergraduate computer programming course in Nigeria (N = 855, an analytic sample derived from 1889 enrolled students).= Statistical analyses, including correlation and regression, examined relationships between ChatGPT usage, student perceptions (enjoyment, confidence, motivation, usefulness), and final scores. Findings reveal a significant divergence: students reported positive perceptions of ChatGPT in terms of affective benefits (e.g., increased confidence and motivation) and task-specific utility (e.g., debugging assistance); however, these perceptions did not correlate with improved academic performance. Regression results indicated that ChatGPT usage and related perceptions had limited predictive power ($R^2 = 0.006$; no predictors were significant). Instructor observations corroborated concerns about over-reliance on AI-generated solutions and potential academic integrity issues. This research underscores the complex, nuanced impact of ChatGPT, highlighting that perceived benefits do not automatically translate into enhanced learning outcomes, and emphasizing the need for careful pedagogical integration strategies. Limitations include the study's single institution design, cross-sectional nature, and reliance on self-report data.

Keywords Academic integrity, ChatGPT, Learning outcomes, Programming education, Student performance

1 Introduction

Programming education has become a cornerstone of higher education, equipping students with essential computational thinking and problem-solving skills that are increasingly relevant across diverse professional fields such as finance, healthcare, and education [15, 85]. As technology continues to evolve, programming competence is no longer limited to computer science disciplines, it has become a critical driver of innovation, employability, and economic growth [33, 38]. Consequently, universities worldwide are emphasizing programming literacy as a fundamental skill necessary for thriving in the digital era.

In recent years, the emergence of generative artificial intelligence (AI) tools, particularly ChatGPT, has introduced new opportunities for transforming programming education. These tools leverage natural language processing to provide adaptive, real-time feedback and personalized learning experiences [4, 93, 99]. Within programming contexts, ChatGPT assists learners by generating code examples, clarifying complex syntax and logic errors, and suggesting alternative approaches to problem-solving [21, 27, 101]. As a result, ChatGPT has become a valuable supplementary learning resource that enhances student engagement and facilitates deeper interaction with programming concepts.

Despite these opportunities, programming education continues to face persistent challenges such as limited access to technical resources, difficulty mastering foundational programming principles, and insufficient immediate feedback [30, 31, 50, 94]. Although AI-driven systems like ChatGPT can help address some of these barriers, scholars have raised important concerns regarding their pedagogical implications. Overreliance on AI tools may hinder the development of students' critical thinking and independent problem-solving abilities, leading to a superficial understanding of programming concepts [40, 82, 87]. Furthermore, empirical evidence on the long-term impact of ChatGPT on programming competence, conceptual understanding, and sustainable learning outcomes remains limited [2, 102].

To address these gaps, this study investigates the impact of ChatGPT on programming education, focusing on students' learning experiences, academic performance, and skill development. Specifically, it seeks to answer the following research questions:

1. How do students perceive ChatGPT as a learning resource in programming courses?
2. What impact does ChatGPT have on students' final programming scores and their perceptions of skill development?
3. How do instructors perceive ChatGPT's influence on students' understanding of programming concepts, debugging abilities, and problem-solving skills?

The significance of this research lies in its potential to guide the effective and responsible integration of AI tools like ChatGPT into programming education. As institutions increasingly adopt AI-supported learning environments, it is essential to understand how these technologies influence student engagement, cognitive development, and academic integrity. Insights from this study will inform educators, administrators, and policymakers on how to balance technological assistance with the cultivation of independent learning and critical thinking.

The other section of this paper is structured as follows. Section 2 presents a comprehensive review of the literature on AI in education and programming pedagogy.

Section 3 outlines the mixed-methods approach employed in this study, integrating quantitative performance metrics with qualitative insights from both students and instructors. Section 4 discusses the findings in relation to the research questions, while Sect. 5 highlights the study's strengths and limitations. Section 6 presents the recommendations for future work. Finally, Sect. 7 presents the conclusions and offers recommendations for future research on AI-enhanced learning environments.

2 Literature review

The integration of ChatGPT into computer programming education has demonstrated significant potential for enhancing student learning experiences and performance through its code generation, debugging, and optimization capabilities. While these features offer valuable support for both students and educators, concerns regarding academic integrity and the accuracy of generated content must be addressed. In this section, we examined the benefits of ChatGPT, reviewed empirical evidence of its effectiveness, and discussed the associated challenges and limitations, highlighting the importance of understanding AI tools in educational contexts.

2.1 Context of AI in education

Artificial Intelligence (AI) refers to technologies that simulate human cognitive functions such as learning, reasoning, and problem-solving [23]. In higher education, AI is increasingly shaping how knowledge is delivered and acquired. It enhances teaching methods, supports data-driven decision-making, and equips students with the skills needed to thrive in a rapidly evolving digital workforce [23, 61]. Tools such as IntelliCode [95], Copilot [16], AlphaDev [55], Codex [24], and DeepMind's AlphaCode [48] demonstrate AI's growing capacity to facilitate learning across multiple disciplines. Within educational contexts, these technologies are applied in adaptive learning systems, intelligent tutoring, and even creative domains such as art education, providing more accessible, personalized, and interactive learning experiences [81].

AI's application in education has also been explored from sociocultural and developmental perspectives. Opesemowo [69] highlights that AI-driven educational systems can bridge learning gaps among diverse communities by enabling contextually inclusive learning experiences. Similarly, Opesemowo and Adekomaya [71] emphasize that integrating AI within higher education not only enhances teaching and learning but also contributes to achieving sustainable development goals by promoting equity, access, and innovation in resource-limited environments. These perspectives underscore the transformative potential of AI in addressing both pedagogical and societal challenges.

A key strength of AI in education lies in its ability to personalize learning. Generative AI tools such as ChatGPT can tailor explanations to individual learners' needs, provide real-time feedback, and promote active engagement. This adaptability is particularly valuable in programming education, where students often struggle with complex syntax or debugging logical errors. Empirical studies indicate that incorporating AI tools into programming coursessuch as Python, Java, and PHP, enhances students' self-efficacy, supports inquiry-based learning, and fosters computational thinking [6, 41, 64].

In the Nigerian higher education context, the integration of AI presents a promising opportunity to strengthen programming education. Generative AI can offer individualized support to students, particularly in resource-limited environments where access to qualified instructors and learning materials may be constrained [66]. Okoye and Mante [65] emphasize that such tools can help maximize student potential by maintaining a balance between guidance and challenge. However, despite these opportunities, awareness and adoption of AI technologies within Nigerian universities remain relatively low, especially among educators [9, 84]. Similarly, Opesemowo and Ndlovu [72] discuss both the positive and problematic dimensions of AI in mathematics education, noting that while AI can facilitate deeper conceptual understanding, it may also diminish students' cognitive engagement if not properly managed.

Nevertheless, the integration of AI in education raises important ethical and pedagogical concerns, particularly surrounding academic integrity. Tools like ChatGPT can assist students in completing assignments in ways that blur the boundary between legitimate assistance and academic dishonesty. Researchers such as Abubakar et al. [3] and Jiao et al. [39] stress the importance of redesigning assessments to ensure that learners genuinely understand the material rather than relying excessively on AI-generated responses. While AI can enhance learning efficiency through instant feedback and clearer explanations, it also risks undermining the development of independent problem-solving and critical-thinking skills.

Evaluations of specific AI tools in programming education have yielded mixed results. For instance, GitHub Copilot has been found capable of solving nearly half of CS1-level problems on its first attempt, with performance improving when prompts are refined in natural language [26]. Comparative studies of ChatGPT, Gemini, AlphaCode, and Copilot reveal notable differences in natural language processing and code generation accuracy across programming languages such as Java, Python, and C++ [86]. These findings suggest that while AI tools hold great promise, their effectiveness largely depends on thoughtful implementation, pedagogical context, and user proficiency.

Beyond the classroom, AI is also transforming educational administration and institutional management. AI-powered data analysis enables universities to make informed decisions, optimize operations, and deliver more personalized student services [56].

Overall, while students and educators alike recognize the transformative potential of AI, particularly in programming and data-driven disciplines, the risks of overreliance, skill erosion, and breaches of academic integrity remain pressing concerns. The challenge, therefore, lies not merely in adopting AI technologies but in integrating them thoughtfully and ethically, ensuring they enhance rather than replace essential human learning processes [35, 98, 105, 107]. Beyond academic integrity, the integration of AI tools in education necessitates a focus on privacy and security, recent research has demonstrated the utility of LLM frameworks in identifying hidden security concerns and privacy violations within software ecosystems [58].

2.2 ChatGPT as a programming education tool

Integrating AI tools, particularly chatbots like ChatGPT, can potentially revolutionize programming education [34]. These AI-driven platforms provide real-time support, enabling students to effectively engage with programming concepts. According to Akpan et al. [5] and Boxleitner [18], these technologies enhance teaching and learning by facilitating personalized learning experiences, aiding in the generation of instructional materials, and providing immediate feedback to learners. ChatGPT, developed by OpenAI, is an advanced AI-powered chatbot utilizing the Generative Pre-trained Transformer

(GPT) model. This model is trained on diverse datasets, enabling it to perform a wide range of tasks, including answering questions, generating code, and providing explanations for complex programming concepts [68]. The primary features of ChatGPT include its ability to engage in natural language conversations, generate coherent and contextually relevant responses, and adapt to the user's level of understanding [11, 13, 91]. This adaptability makes ChatGPT a valuable tool for students seeking assistance in programming.

Recent findings reinforce these benefits. Opesemowo et al. [70] reported that educators increasingly perceive ChatGPT as an effective instructional aid, capable of streamlining grading and feedback, though concerns about content reliability and the risk of overdependence remain. ChatGPT can assist with various tasks, including content creation, information retrieval, translation, and tutoring, while its architecture enables it to adapt across different domains [83]. AI chatbots in education have been shown to enhance student engagement and motivation, as they offer immediate assistance and foster a more interactive learning environment [101, 103]. However, while these tools present significant opportunities, they also raise concerns regarding academic integrity and the development of critical thinking skills [1, 20, 28]. Moreover, while ChatGPT assists beginners with syntax errors, recent advancements in automated verification suggest a move toward cross-multimodal systems that integrate developer responses and visual analysis to validate complex bug fixes [57].

Opesemowo, Abanikannda, and Iwintolu [70] further explored lecturers' perceptions of ChatGPT for instructional assessment, revealing that while educators recognize its potential to enhance teaching efficiency and feedback quality, concerns persist regarding accuracy, ethical use, and overreliance. These insights are crucial for contextualizing AI adoption in developing educational systems such as Nigeria's. Notably, ChatGPT offers several capabilities that enhance the programming learning experience. One of its most significant features is code generation, which enables the production of code snippets based on user prompts. This functionality allows students to quickly generate examples and solutions, facilitating a deeper understanding of programming concepts [87, 101, 102]. Additionally, ChatGPT provides debugging assistance by helping students identify and resolve errors in their code. This support is particularly beneficial for beginners who may struggle with common programming pitfalls [46, 76]. Additionally, ChatGPT can provide personalized learning pathways by assessing a student's existing knowledge and recommending relevant resources, tutorials, and exercises tailored to their specific needs [90, 101]. This personalized approach fosters a more engaging and effective learning environment [99]. Its integration has been met with generally positive feedback, and studies indicate significant improvements in learning outcomes, particularly in self-directed environments and among non-computer science majors [44, 73].

As shown in Fig. 1, ChatGPT's capabilities in code generation, debugging assistance, and personalized support make it a valuable resource for students. However, the effective use of ChatGPT requires guidelines to maintain academic integrity and avoid reliance on AI tools for foundational skills [79]. Innovations like GPTutor [21, 22], PyTutor [100] demonstrate the potential for personalized tutoring in programming, addressing individual learning needs. Furthermore, the combination of pair programming techniques with ChatGPT suggests a promising approach to overcoming traditional challenges in programming education [12].

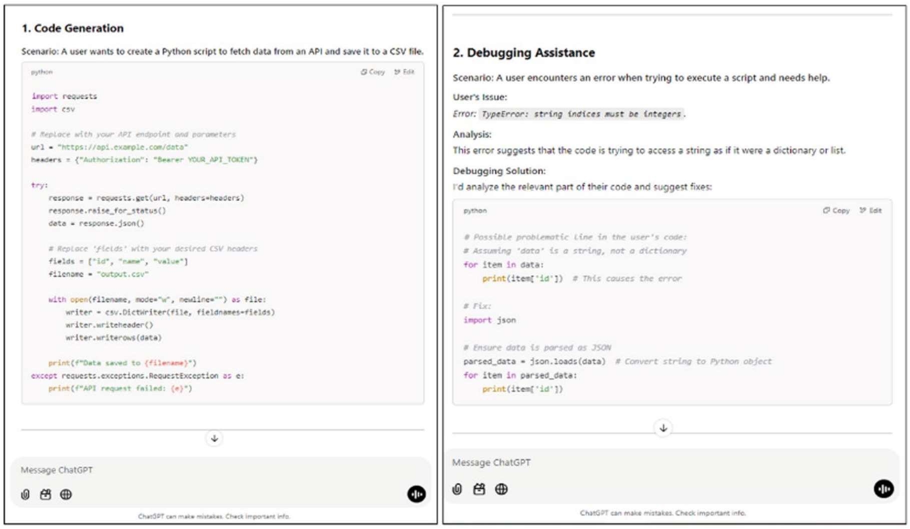
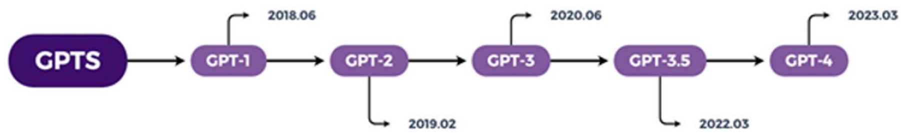


Fig. 1 Samples of code generation, debugging assistance on ChatGPT

Several studies have explored the integration of ChatGPT in programming education, web programming courses, or computer science, highlighting its impact on student engagement and performance [34, 46, 60, 87, 96, 99]. Research indicates that students who utilize ChatGPT report higher levels of engagement and satisfaction with their learning experience. For instance, a study by Abdulla et al. [2] found that students using ChatGPT demonstrated improved problem-solving skills and a greater willingness to explore programming concepts independently. Research indicates that students who utilize ChatGPT as a supplementary resource demonstrate increased motivation and participation in programming tasks [40, 87]. Rane [80] noted that ChatGPT enhances engagement and provides timely feedback, yet emphasizes the challenges of emotional intelligence and learning experiences personalization. Alan and Yurt [7] found that incorporating ChatGPT into flipped classrooms significantly improved student interaction and satisfaction. Ma et al. [52] reported similar findings in programming education, noting that students appreciated ChatGPT for debugging and task assistance, but highlighted concerns about over-reliance and the quality of AI feedback. Maher et al. [53] and Aviv et al. [8] investigated the impact of ChatGPT on student learning outcomes by measuring and analyzing participants' learning outcomes through their UML diagrams and programming. Similarly, Maher et al. [54] investigated the impact of ChatGPT on student engagement and performance in introductory Java programming courses using a mixed-methods research approach. Lastly, Onofre et al. [67] emphasized ChatGPT's role in fostering collaboration between professors and students, enhancing personalized learning experiences.

By offering a range of features that enhance teaching and learning, ChatGPT has evolved through several iterations: GPT-1, GPT-2, GPT-3, and the latest, GPT-4. As shown in Fig. 2, each version is built upon previous developments, leading to increasingly sophisticated AI capabilities. GPT-1 laid the foundation, while GPT-2, launched in 2019, significantly improved performance through a larger model and broader dataset. GPT-3, released in 2020, features 175 billion parameters, enhancing generalization across various tasks [78, 106]. ChatGPT, particularly in its iterations GPT-3.5 and GPT-4 used in this study, serves as a transformative tool in programming education by



GPT Models	GPT-1	GPT-2	GPT-3	GPT-3.5	GPT-4
Parameters (10^9)	0.117	1.5	175	1.3	Official report unavailable
Dataset	BooksCorpus (over 40GB)	WebText (40TB)	Common Crawl (45TB)	Common Crawl (45TB)	Official report unavailable

Fig. 2 GPTs Development Timeline [106]

acting as a personalized tutor, educator assistant [75]. The architecture of these models relies on the generative pre-trained transformer framework, leveraging deep learning to process and generate human-like text based on extensive datasets [45]. Through systematic evaluations, these models have demonstrated capabilities in various educational scenarios, such as program code instruction, hint generation, grading feedback, peer programming, task creation, and contextualized explanations [88]. Kürger and Gref [47] evaluated the performance of ChatGPT-3.5 and GPT-4.0 in the context of an undergraduate computer science program. They found that ChatGPT-3.5 achieved 79.9% of the maximum possible scores across 10 tested modules, while BingAI scored 68.4%, and the 65-billion-parameter variant of LLaMa scored 20%. As an educational tool, ChatGPT exemplifies the integration of advanced AI in fostering programming skills among learners.

2.3 Impact of ChatGPT on learning outcomes

ChatGPT can positively impact students' learning experiences in programming courses and other educational contexts. Studies indicate that the use of AI tools, including ChatGPT, can enhance student engagement and facilitate personalized learning by allowing students to explore programming concepts at their own pace [35, 53]. Students generally report improved problem-solving and writing abilities, though concerns about over-dependence and its effects on critical thinking persist [43]. For instance, a study by Logozar et al. [49] conducted a rigorous evaluation of ChatGPT's performance in C and C + + programming assignments. The study revealed that ChatGPT consistently produced high-quality solutions that surpassed typical student performance, particularly in adapting solutions to changing requirements. However, this effectiveness varies by application area. This finding aligns with Yilmaz and Yilmaz's [102] work, which reported positive correlations between ChatGPT usage and improved computational thinking skills among students. These studies collectively suggest that ChatGPT serves as a valuable tool in facilitating deeper engagement with programming material, ultimately leading to better academic outcomes. Jalil et al. [37] found more modest results in software testing education, with ChatGPT achieving only 37.5% accuracy in answering technical questions, suggesting limitations in specialized programming domains. These studies collectively suggest that ChatGPT serves as a valuable tool in facilitating deeper engagement with programming material, ultimately leading to better academic outcomes.

Qualitative feedback from students further illuminates their experiences using ChatGPT in programming tasks. Many students reported that ChatGPT provided immediate assistance and clarification on complex topics, significantly reducing their frustration and enhancing their learning experience. For example, participants in a study by Biswas [17] noted that ChatGPT helped them grasp difficult computer programming concepts more effectively than traditional methods. However, some students raised concerns about over-reliance on the tool, fearing it might hinder their ability to think critically and solve problems independently. This duality in student feedback highlights the necessity for educators to implement ChatGPT thoughtfully, ensuring that it complements rather than replaces traditional learning strategies. A survey conducted by Murad et al. [62] revealed that students appreciated the tool's ability to generate code snippets and offer debugging suggestions, which facilitated their learning process. Also, instructor concerns regarding the reliability of AI-generated code are supported by research into user-reported 'hallucinations,' where LLMs provide confidently presented but factually incorrect information [59]. However, students also expressed concerns about becoming overly dependent on the tool, fearing that it might hinder their ability to think critically and solve problems independently. Pardos and Bhandari [74] found that 70% of ChatGPT-generated hints in programming-related mathematical problems led to positive learning outcomes. However, Basic et al.'s [14] controlled study revealed that students working without ChatGPT outperformed those using it in certain assessment criteria, highlighting the importance of proper integration strategies. Moreover, qualitative findings suggest that integrating AI tools in programming courses can enhance learning outcomes, foster teamwork, and boost self-efficacy among students [19]. Therefore, while ChatGPT offers significant benefits, careful integration and balancing its usage with traditional learning methods are crucial for maximizing its educational value [87].

However, several studies have highlighted important limitations in ChatGPT's educational application. For instance, Rahman and Watanobe [77] found that students who used ChatGPT demonstrated improved debugging skills, however, the tool's overall problem-solving capabilities remained limited, particularly in more complex tasks. Also, Azaria et al. [10] documented instances of inaccurate code generation by ChatGPT, along with other errors, emphasizing the need for careful oversight when using the tool in programming education.

Despite these limitations, some studies have also highlighted positive outcomes of ChatGPT's use. Research by Tian et al. [97] highlighted that students who engaged with ChatGPT reported improved comprehension of programming syntax and logic. The tool's ability to provide instant feedback and explanations contributed to a more interactive learning experience, allowing students to grasp difficult concepts more effectively. Similarly, Hu et al. [36] found that students who engaged with ChatGPT demonstrated a greater ability to analyze and evaluate programming problems, suggesting that the tool encourages higher-order thinking. Additionally, the study by Ghimire and Edwards [32] indicated that students reported increased self-efficacy in their programming abilities when using ChatGPT, as they felt more confident in tackling challenging assignments. While these findings demonstrate ChatGPT's potential, they underscore the need for careful integration into educational practices, ensuring that the tool complements traditional learning methods without fostering over-reliance.

Despite the promising results, addressing the limitations and challenges associated with ChatGPT's integration into programming education is crucial. Some studies, such as those by Jošt et al. [40] and Sok and Heng [89], highlighted that reliance on ChatGPT could lead to superficial learning, where students may obtain solutions without fully understanding the underlying concepts. Furthermore, concerns regarding the accuracy of ChatGPT's outputs and the potential for generating incorrect code must be acknowledged. As educators explore the integration of ChatGPT, it is essential to establish guidelines that promote responsible usage, ensuring that students leverage the tool to enhance their learning while maintaining academic integrity.

2.4 Research gaps and motivation

While existing research highlights the potential of ChatGPT in programming education, significant gaps remain regarding its long-term impact on student learning. Studies have shown that ChatGPT can enhance code generation and debugging [42] and increase student engagement [27]. However, there is still a limited understanding of how sustained use affects critical thinking and problem-solving abilities over time. It remains unclear whether ChatGPT fosters a deeper understanding of programming concepts or merely supports surface-level learning, where students rely on the tool for solutions without fully grasping the underlying material [3, 39].

In addition, although prior studies have demonstrated ChatGPT's effectiveness, there is a noticeable lack of research exploring students' perceptions of the tool. Chaudhry et al. [20] emphasize that understanding how students perceive ChatGPT, whether as a supportive learning resource or as a shortcut, could provide valuable insights into its influence on learning autonomy. Such understanding is crucial to ensure that ChatGPT encourages active learning rather than fostering passive dependence.

Another gap concerns ChatGPT's performance across different programming tasks. While it performs well in general coding contexts, its effectiveness in assisting with more complex or specialized programming challenges has not been thoroughly investigated [86, 103].

Therefore, this study aims to address these research gaps by investigating the impact of ChatGPT on student engagement, problem-solving skills, and long-term skill development. The findings aim to provide deeper insights into how AI tools can complement traditional teaching methods while preserving critical thinking and independent learning.

3 Research methodology

This study adopted a mixed-methods approach to examine the impact of ChatGPT on students' learning in programming courses. The research employed a combination of quantitative and qualitative methods to provide a comprehensive understanding of students' experiences and perceptions.

The quantitative aspect involved a survey to assess students' engagement, motivation, and confidence when using ChatGPT. The survey also collected data on students' prior experience, their ability to learn new concepts, and the impact of ChatGPT on their programming skills and problem-solving abilities. Data were analyzed using descriptive statistics and multiple regression to identify correlations between ChatGPT use and improvements in learning outcomes.

The qualitative aspect involved semi-structured interviews with students to gather in-depth insights into their challenges and experiences using ChatGPT. Thematic analysis was used to identify recurring themes related to effectiveness and student perceptions.

The study aimed to answer three research questions: How do students perceive the role of ChatGPT in learning? What impact does it have on their programming skills and learning outcomes? How do instructors view its effect on students' abilities?

3.1 Research design

The study used a cross-sectional research design, supported by instructor observations and students' open-ended responses, to explore how ChatGPT influences teaching and learning in programming courses. This design choice was driven by the need to collect large-scale quantitative data on students' experiences, while also gaining rich qualitative insights into learning outcomes and behavioral changes. The survey method allowed for efficient data collection from a large student population, while ensuring methodological rigor through validated instruments and systematic analysis procedures. Figure 3 presents an overview of the research methodological flow.

The research incorporated three primary data collection methods: First, an online survey was administered to students to gather information about their usage patterns, perceptions, and experiences with ChatGPT in programming courses. Second, instructor feedback was collected to provide a professional assessment of student learning outcomes and behavioral changes. Finally, course performance data was analyzed to explore the relationship between ChatGPT usage and academic achievement.

3.2 Participants

This study was conducted among undergraduate students enrolled in computer programming-related courses across four faculties—Science, Technology, Environmental Design and Management, and Education—at Obafemi Awolowo University, Nigeria, during the second semester of the 2023/2024 academic year. The programming instruction in these courses was based on the Python Programming Language (PPL).

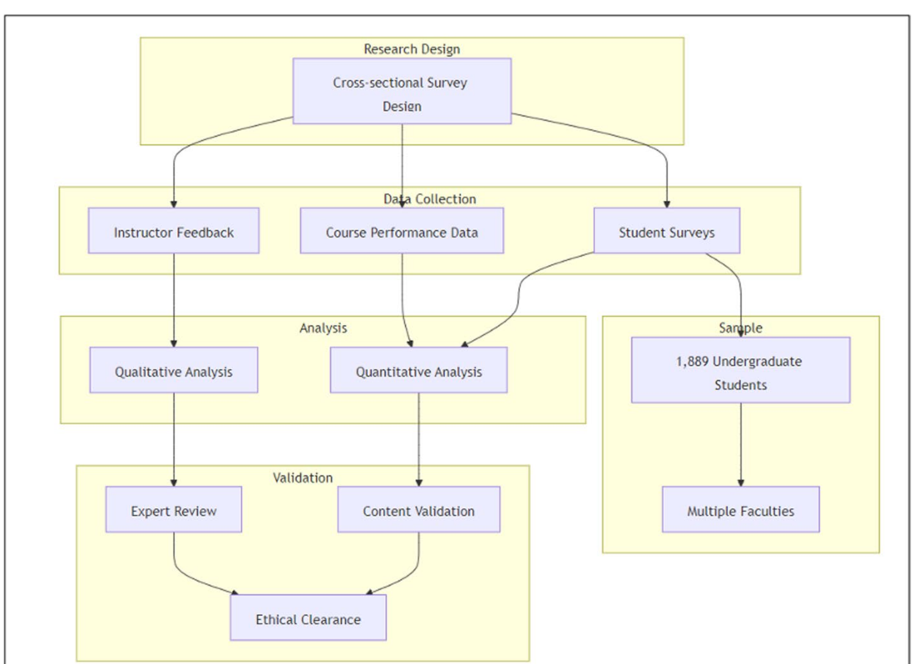


Fig. 3 Research Methodology Flow

A total of 1889 students were invited to participate in the research. Male and female students were surveyed. Following approval by the institution's ethical committee. Of these, 1200 students responded to the online survey, representing a response rate of 63.5%. Following data cleaning to remove incomplete or partially filled questionnaires, 855 valid responses were retained for analysis, yielding a completion rate of 71.3%. Exclusions were made primarily due to missing demographic information and incomplete answers to key survey sections.

Participants were distributed across faculties as follows: 40.9% from the Faculty of Science, 38.8% from the Faculty of Technology, 11.9% from the Faculty of Environmental Design and Management, and 8.4% from the Faculty of Education. A survey was conducted with 17 instructors to gather their insights, which are invaluable for understanding how students' use of ChatGPT affects their learning and performance. This diverse representation ensured a balanced and comprehensive view of students' experiences and perceptions regarding the educational use of ChatGPT in programming contexts.

3.3 Method

This study sought to examine students' prior usage of ChatGPT and their familiarity with OpenAI's software to better understand its perceived impact on their learning in programming courses. Survey responses revealed that 89.7% of students had previously used ChatGPT, with 87.1% reporting that they used it primarily for academic purposes. About 8.1% of students indicated no prior usage, while a small percentage responded with "Maybe."

All students had completed foundational theoretical training in Python programming during the first semester, establishing a baseline understanding of essential programming concepts. The second semester, however, was more practice-oriented, focusing on developing students' applied programming skills. Students engaged in a series of assignments requiring coding solutions to problems of varying complexity, covering key areas such as files, dictionaries, lists, tuples, loops, and object orientation. They also participated in lab sessions where they tackled specific problem-solving challenges to reinforce their practical understanding of Python.

An online survey was administered via Google Forms at the end of the semester to collect data on students' experiences, perceptions, and engagement with ChatGPT, which consisted of 40 items. In-class encouragement was given to promote survey completion, but the survey itself was unsupervised and not time-restricted, allowing students to complete it independently and at their convenience; however, it is estimated to take approximately 15–20 min to complete.

In addition to surveying students, a version of the survey was also administered to the course coordinator (instructor) and other lecturers in the teaching and practical team. This was different from the student survey as it focuses on capturing instructors' observations regarding the impact of ChatGPT on students' learning outcomes and problem-solving skills. The inclusion of the instructor's perspective helped provide a balanced view, considering both students' self-reported experiences and the instructor's professional assessment. The instructors were 17 with knowledge on AI and basic coding, and they oversaw both semesters, using consistent instructional strategies, materials, and assessment methods to maintain a uniform learning environment across the cohort.

This consistency allowed the study to reliably assess the influence of ChatGPT on student outcomes by minimizing instructional variability.

After completing coursework and lab sessions, the instructor also provided feedback to students, helping evaluate their grasp of programming concepts and their ability to apply them in practice. Combining students' and instructors' insights, this dual perspective was intended to yield a comprehensive understanding of ChatGPT's role in the educational setting.

3.4 Data collection and handling

A census sampling approach was adopted, which is a method that involves collecting data from every member of the target population rather than selecting a subset. For data collection, a detailed online questionnaire was administered, as described earlier. The survey was distributed to all 1889 students enrolled in the programming course. A total of 1200 students responded, representing a response rate of 63.5%. After data cleaning, which involved removing incomplete or inconsistent responses to ensure analytical accuracy, a final dataset of 855 complete responses was retained for analysis. This corresponds to a completion rate of 71.3% among those who started the survey.

Incomplete responses were excluded primarily due to missing answers in key sections related to programming performance or perceptions of ChatGPT's usefulness. The data cleaning process also involved checking for duplicate entries and ensuring consistent coding across all variables.

Ethical approval was obtained from the university's review committee prior to data collection, ensuring full compliance with institutional guidelines on research involving human participants. The survey was voluntary and anonymous, and informed consent was obtained from all participants before participation.

The final dataset, comprising 855 complete cases, provided a reliable foundation for both the quantitative and qualitative analyses conducted in the study.

3.5 Survey instrumentation and response analysis

The survey was structured to capture a comprehensive set of responses through both categorical and Likert scale questions, designed to gather insights into participant demographics, educational backgrounds, and programming proficiency. Key demographic breakdowns are presented in Table 1. For each demography, we provided an explanation of the analysis, as shown in the remarks of Table 1.

Table 1 Key demographic breakdown and response analysis

SN	Demographics	Remarks
1	Age Group	Participants' ages were categorized into six brackets, coded from 1 to 5 (with 1 representing the youngest bracket, "0–17," and 5 the oldest, "42 and above"). Approximately 85.3% of respondents fell within the "18–23" bracket (coded as 2), indicating a predominantly young demographic
2	Gender Identity	Respondents identified as either male or female, with 61.7% identifying as male and 38.0% as female, suggesting a balanced gender distribution within the binary options provided
3	Faculty Level	Participants' academic backgrounds spanned various disciplines, primarily within the Engineering, Education, and Science faculties.
4	Programming Experience	Responses on programming experience were categorized as "Yes," "No," or "Maybe." Approximately 51.6% of respondents indicated "No," while 45.3% responded "Yes." These results suggest that despite prior exposure to programming courses, a significant proportion of respondents still lacked practical coding proficiency

3.6 Survey instrumentation

The survey instrument was divided into 9 sections, with a total of 40 items. The sections were as follows:

1. Demographics–5 items
2. Experience with Programming–2 items
3. Experience with ChatGPT–3 items
4. Perception of ChatGPT as a Learning Resource–4 items
5. Comparison with Other Learning Resources–3 items
6. Motivation and Engagement–5 items
7. Learning Outcomes–13 items
8. General Feedback–5 items

Each section targeted a specific aspect of ChatGPT usage, such as frequency of use, perceived effectiveness, impact on programming skills, and challenges encountered. The instrument was grounded in existing literature on educational technology, AI tools in learning, and the use of the Likert scale for structuring questions. These items were designed to capture both qualitative and quantitative data on students' experiences and outcomes related to ChatGPT's integration into their learning process. Also, the Likert scale rating, for example, was 1-Strongly agree, 2-Agree, 3-Neutral, 4-Disagree, 5-Strongly Disagree.

3.6.1 Survey design

The survey was organized into nine sections, comprising 40 items, including open-ended and closed-ended questions to ensure comprehensive coverage of possible key factors related to students' engagement with ChatGPT as a learning tool. These sections were strategically chosen based on the research objectives to capture various dimensions, including students' frequency of use, their perceptions of its effectiveness, and its impact on programming skills this structure provided a comprehensive overview of ChatGPT's impact on students' learning outcomes and performance in programming courses. Each section targeted specific aspects of ChatGPT usage, allowing us to capture detailed insights across various dimensions.

Most closed-ended questions employed a 5-point Likert scale with variations such as "Strongly Disagree" to "Strongly Agree," "Always" to "Never," "Very Confident" to "Not Very Confident," "Very Reliable" to "Not Very Reliable," "Very Satisfied" to "Very Dissatisfied," and "Very Important" to "Not Very Important." These variations enabled precise measurement across different response types, reflecting the range and intensity of students' perceptions toward ChatGPT as a learning tool.

The Likert scale was instrumental in assessing ChatGPT's influence on factors like motivation, confidence, problem-solving abilities, and engagement. In addition to quantitative data, open-ended questions in each section gathered qualitative insights from students and instructors, providing nuanced information on specific experiences and challenges with ChatGPT.

The nine-section structure, paired with a mixed-methods approach, strengthened the survey's reliability and validity, giving a well-rounded view of ChatGPT's role in programming education by capturing both quantifiable attitudes and unique perspectives from individual respondents.

3.6.2 Validity and reliability

Prior to launching the survey, it underwent a comprehensive validation process to ensure alignment with the study's research objectives. The research team, which includes educational specialists and researchers with a clear understanding of AI, critically reviewed and refined the survey content, ensuring clarity, relevance, and coverage of key areas necessary to achieve the study's aims. Each question was properly structured, with the survey items crafted to capture constructs central to understanding ChatGPT's impact on programming learning outcomes. This approach supported by similar study by Youssef et al. [104] on the impact of chatgpt on academic learning.

To further strengthen content validity, we followed best practices similar to the sequential mixed-method approach, which emphasizes collaborative development between content experts and assessment specialists. This method is supported by established methodologies in survey validation literature, including the structured processes described by key scholars such as Cohen, Manion and Morrison [25]. Additionally, in line with best practices for assessments in vocational and educational contexts, expert reviewers were engaged to confirm that survey items effectively captured relevant dimensions of programming knowledge and learning behaviours.

3.7 Statistical analysis methodology

The statistical analysis aimed to rigorously explore the relationships between students' programming scores and their perceptions of ChatGPT as a learning resource. This phase of the research was carried out using Ordinary Least Squares (OLS) multiple regression on the RegressIt Package. The purpose was to understand how various factors influenced by ChatGPT, such as problem-solving skills, debugging capabilities, and understanding of programming concepts, impacted learning outcomes. The analysis strictly adhered to the assumptions of OLS regression, such as linearity, independence, homoscedasticity, normality of residuals, and no multicollinearity, to ensure reliable and valid results, reflecting the professional standards expected of MBA-level research [29].

3.7.1 Objective of the statistical analysis

The regression analysis sought to answer three key research objectives. First, it examined how students perceive ChatGPT as a resource, particularly its influence on enjoyment, confidence, motivation, prior programming experience, and the acquisition of new programming concepts. Second, it evaluated the impact of ChatGPT on programming skills and learning outcomes, including its utility for completing assignments, understanding concepts, improving problem-solving abilities, and contributing to course success. Lastly, it investigated markers' perceptions of ChatGPT's influence on students' programming skills, focusing on their understanding of concepts, debugging abilities, and problem-solving approaches. These objectives framed the analytical approach and guided the interpretation of results. The overarching goal of the analysis was to assess the utility of ChatGPT as a pedagogical tool and its broader implications for programming education.

3.7.2 Descriptive statistics

The initial phase of the statistical analysis involved exploring the dataset using descriptive statistics to identify underlying trends and variability. Key metrics such as the mean, median, standard deviation (SD), and range were computed for all relevant variables. For example, students' programming scores, measured on a 0–10 scale, had a mean score of 7.213, with a standard deviation of 3.566, indicating moderate variability in student performance.

Responses to Likert-scale questions, assessing factors such as ChatGPT's perceived effectiveness, were similarly analyzed. For instance, the mean score for the item "How important is the availability of ChatGPT to your success in the course?" was 3.662 (SD = 0.822) on a 5-point scale, with a median of 4. This reflects that students generally considered ChatGPT an important resource for their academic success. Variability in responses was captured using the standard deviation, which showed the degree of divergence in student opinions.

These descriptive statistics provided a solid foundation for further analysis, highlighting key trends and areas of interest. The results enabled the identification of variables that were most relevant to the research objectives, guiding subsequent statistical methods. This approach ensured that the analysis remained focused on the factors influencing students' engagement with ChatGPT in programming courses.

3.7.3 Correlation analysis

Correlation analysis was performed to explore relationships between students' programming scores and their perceptions of ChatGPT's utility. The Pearson correlation coefficients indicate statistically significant positive associations between programming scores and variables such as ChatGPT's role in understanding programming concepts and debugging code. These findings established a foundation for the regression model, highlighting which factors were most likely to influence students' outcomes.

3.7.4 Multiple regression analysis using OLS

Ordinary Least Squares (OLS) multiple regression analysis was employed to examine the relationship between students' programming scores and key factors that reflected their perceptions of ChatGPT's usefulness in programming education. This method was chosen for its ability to model linear relationships between one dependent variable (students' programming scores) and several independent variables, including perceived usefulness for problem-solving, debugging, assignment completion, and overall course success. The analysis was designed to provide a quantitative assessment of ChatGPT's influence and to address the study's main research objectives: (i) to explore how students perceive ChatGPT as a learning resource in terms of confidence, motivation, and understanding of programming concepts, (ii) to evaluate the tool's impact on students' programming skills and overall performance, and (iii) to consider instructors' views on how ChatGPT affects students' understanding, debugging, and problem-solving abilities.

Although the study's variables were based on Likert-type scales, the composite scores generated from multiple items were treated as continuous variables for statistical analysis. This approach is consistent with prior research in education and social sciences, where Likert-scale data are aggregated to approximate interval-level measurement suitable for regression analysis [63, 92]. Ordinary Least Squares (OLS) regression was therefore appropriate for examining predictive relationships among variables, allowing for simultaneous testing of multiple factors and yielding interpretable coefficients aligned with the study's objectives.

Prior to conducting the regression analysis, diagnostic checks were carried out to confirm that the core assumptions of Ordinary Least Squares (OLS) regression were reasonably satisfied. Linearity between predictors and the dependent variable was visually assessed through scatterplots, which showed no major deviations from a linear pattern. Multicollinearity was examined using the Variance Inflation Factor (VIF), with all predictors recording values below 5, indicating no significant intercorrelation among independent variables.

The regression results indicated that none of the predictors were statistically significant at the 5% level. The model produced an R^2 value of 0.006 and an adjusted R^2 of 0.002, meaning that the independent variables collectively explained less than 1% of the variance in students' programming scores. These results suggest that students' perceptions of ChatGPT, such as its usefulness for debugging, understanding programming concepts, or completing assignments, did not significantly predict their academic performance. The Root Mean Square Error (RMSE) of 3.457 further indicated a substantial amount of unexplained variability in the model.

Overall, while students may perceive ChatGPT as a valuable learning aid, these findings show that their perceptions and reported usage do not directly translate into measurable improvements in programming scores. This suggests that other factors, such as prior programming experience, study habits, and instructional methods, may play a more prominent role in influencing student performance outcomes.

3.7.5 Interpretation in relation to research objectives

The regression analysis provided valuable insights into how ChatGPT impacts students' programming skills and learning outcomes. The analysis addressed the second research objective by quantifying the importance of factors such as problem-solving, debugging, and understanding programming concepts. It demonstrated that ChatGPT significantly contributes to enhancing students' learning experiences, particularly in critical areas like concept mastery and error correction.

The findings also shed light on students' perceptions of ChatGPT as a learning resource, addressing the first objective. Variables related to confidence, motivation, and prior programming experience emerged as important contributors to students' performance. Furthermore, the analysis highlighted markers' perceptions of ChatGPT's role in shaping problem-solving approaches and debugging abilities, thereby fulfilling the third research objective.

3.7.6 Qualitative data analysis approach

In addition to the quantitative analyses, qualitative data were collected from students' open-ended survey responses and instructors' reflective feedback. These responses provided deeper insight into students' experiences with ChatGPT and the instructors' observations of how the tool influenced students' programming learning.

The qualitative data were analyzed using a thematic coding approach. Responses were first reviewed to identify recurring ideas related to how students used ChatGPT, the types of support it provided, and the challenges they encountered. Common themes included concept understanding, problem-solving, debugging assistance, motivation, and concerns about academic integrity. These themes were refined through repeated reading and comparison to ensure consistency and clarity in interpretation.

The instructor feedback was analyzed in the same way, focusing on their observations of students' engagement, performance, and learning behaviors when using ChatGPT. This approach allowed the qualitative data to complement the quantitative findings by providing context and explanations for the patterns observed in the survey results.

3.8 Ethical considerations

Ethical approval for this study was obtained from the university's research ethics committee. Prior to participation, informed consent was obtained from all participants, ensuring they were fully informed about the study's objectives, the voluntary nature of their involvement, and their right to withdraw at any time without penalty. Participants were assured that their responses would be treated confidentially and used exclusively for the purposes of this research. The survey was anonymous, with no identifying information collected. All data was securely stored and accessible only to the research team**.**

4 Results and discussion

4.1 Assessing student perceptions of ChatGPT for programming learning (*RQ1*)

This section presents the results of investigating programming students' perceptions of ChatGPT as a learning resource, specifically examining its impact on their enjoyment, confidence, motivation, the influence of prior experience, and its effectiveness for learning new concepts. Thus, the detailed analysis of ChatGPT's impact on students' programming was based on descriptive analysis, correlation analysis with the usefulness of ChatGPT as the dependent variable, and regression analysis for the usefulness of ChatGPT as a learning resource.

4.1.1 Descriptive analysis on RQ1

In Fig. 4, the dataset evaluates participants' perceptions of ChatGPT as a learning resource for programming. The analysis provides insights into its usefulness, impact on confidence and motivation, enjoyment, and ability to support learning beyond traditional classroom content.

As Fig. 4 reflects, most respondents view ChatGPT as a useful learning resource, with an average usefulness rating of 2.56 on a 3-point scale. The data shows consistency, with a median rating of 3, and moderate variability (SD = 0.582SD = 0.582). The slightly left-skewed distribution (Skewness = – 0.919Skewness = – 0.919) indicates that a small number of respondents found it less useful. Overall, responses are evenly spread, as indicated by a near-normal kurtosis (– 0.150–0.150).

Regarding confidence in programming, participants rated the tool favourably, with an average score of 3.72 on a 5-point scale. The median of 4 suggests that most respondents found ChatGPT to boost their confidence moderately to significantly. The data is symmetrically distributed (Skewness = – 0.091Skewness = – 0.091) with few extreme responses, as supported by a flat kurtosis value (– 0.491–0.491) and moderate variability (SD = 0.791SD = 0.791).

Descriptive Statistics

Variable	# Fitted	Mean	Median	Std.Dev.	Root.M.Sqr.	Std.Err.Mean	Minimum	Maximum	Skewness	Kurtosis
How_would_you_rate_the_usefulness_of_ChatGPT_as_a_learning_resource?	855	2.557	3.000	0.582	2.622	0.020	1.000	3.000	-0.919	-0.150
Do_you_think_ChatGPT_increases_your_confidence_in_your_programming_abilities?	855	3.716	4.000	0.791	3.799	0.027	2.000	5.000	-0.091	-0.491
Do_you_think_ChatGPT_makes_learning_programming_more_enjoyable_for_you?	855	3.935	4.000	0.726	4.001	0.025	2.000	5.000	-0.176	-0.435
Do_you_think_using_ChatGPT_as_a_learning_resource_increases_your_motivation_to_learn_programming?	855	3.747	4.000	0.799	3.832	0.027	2.000	5.000	-0.121	-0.518
Does_ChatGPT_help_you_learn_new_programming_concepts_not_covered_in_class?	855	1.665	2.000	0.472	1.731	0.016	1.000	2.000	-0.703	-1.510
Have_you_used_ChatGPT_before_for_any_purpose?	855	0.984	1.000	0.127	0.992	0.004343	0.000	1.000	-7.635	56.425
How_would_you_rate_your_prior_experience_with_programming_before_taking_the_course?	855	3.379	3.000	0.796	3.471	0.027	2.000	5.000	0.080	-0.450

Fig. 4 Descriptive Analysis of *RQ1*

ChatGPT's ability to make programming more enjoyable received the highest average rating of 3.93, with a median of 4, reflecting widespread positive perceptions. The distribution of responses is nearly symmetrical (Skewness = – 0.176Skewness = – 0.176) and flat (Kurtosis = – 0.435Kurtosis = – 0.435), with minimal variability (SD = 0.726SD = 0.726). This suggests consistent enthusiasm among participants.

In terms of motivation to learn programming, respondents rated ChatGPT at an average of 3.75, with a median of 4, indicating it inspires learners to engage further. The responses are symmetrically distributed (Skewness = – 0.121Skewness = – 0.121) and show a moderate spread (SD = 0.799SD = 0.799). These results suggest that ChatGPT plays a role in driving programming-related motivation.

When asked about its role in supporting learning of concepts not covered in class, most respondents affirmed its usefulness, with a mean score of 1.67 on a binary scale (1 = Yes, 2 = No). The responses are moderately concentrated around "Yes" (Skewness = – 0.703Skewness = – 0.703) and exhibit minimal variability (SD = 0.472SD = 0.472). This suggests that ChatGPT is seen as a valuable supplementary tool.

An overwhelming 98.4% of participants reported prior usage of ChatGPT, as reflected in the binary variable mean of 0.984. The extreme left skew (Skewness = – 7.635Skewness = – 7.635) and peaked distribution (Kurtosis = 56.425Kurtosis = 56.425) confirm that the vast majority had prior familiarity with the tool, leaving minimal variability (SD = 0.127SD = 0.127) in responses.

Lastly, prior programming experience among participants averaged 3.38 on a 5-point scale, with a median of 3. This indicates that most respondents had moderate experience with programming. The symmetrical (Skewness = 0.080Skewness = 0.080) and flat distribution (Kurtosis = – 0.450Kurtosis = – 0.450) suggest that participants' backgrounds are diverse but relatively evenly distributed.

4.1.2 Correlation analysis on RQ1

The correlation analysis highlights significant relationships between the usefulness of ChatGPT as a learning resource (dependent variable) and other factors. Figures 5 and 6 illustrate the interpretation of these correlations in matrix (Fig. 5) and graphical (Fig. 6) forms.

145275302120900011974830212090000As shown in Fig. 6, the usefulness of ChatGPT correlates moderately with *confidence in programming abilities* (r = 0.357r = 0.357), indicating that individuals who find ChatGPT useful are more likely to feel confident in their programming skills. This underscores the tool's role in fostering self-assurance in learners. Similarly, there is a positive correlation with *enjoyment of programming (*r = 0.402r = 0.402**)**, suggesting that those who find ChatGPT useful tend to enjoy the learning process more. This link emphasizes ChatGPT's potential to enhance the overall learning experience.

The relationship between usefulness and *motivation to learn programming* (r = 0.333r = 0.333) is also noteworthy. This moderate positive correlation implies that

Correlation Matrix (n=855)

Variable	How_would_you_rate_the_usefulness_of_ChatGPT_as_a_learning_resource?	Do_you_think_ChatGPT_increases_your_confidence_in_your_programming_abilities?	Do_you_think_ChatGPT_makes_learning_programming_more_enjoyable_for_you	Do_you_think_using_ChatGPT_as_a_learning_resource_increases	Does_ChatGPT_help_you_learn_new_programming_	Have_you_used_ChatGPT_before_for	How_would_you_rate_y
How_would_you_rate_the_usefulness_of_ChatGPT_as_a_learning_resource?	1.000						
Do_you_think_ChatGPT_increases_your_confidence_in_your_programming_abilities?	0.357	1.000					
Do_you_think_ChatGPT_makes_learning_programming_more_enjoyable_for_you?	0.402	0.620	1.000				
Do_you_think_using_ChatGPT_as_a_learning_resource_increases_your_motivation_to_learn_programming?	0.333	0.657	0.542	1.000			
Does_ChatGPT_help_you_learn_new_programming_concepts_not_covered_in_class?	0.210	0.291	0.305	0.275	1.000		
Have_you_used_ChatGPT_before_for_any_purpose?	0.092	0.000	0.052	0.005	0.026	1.000	
How_would_you_rate_your_prior_experience_with_programming_before_taking_the_course?	0.090	0.229	0.221	0.221	0.185	0.015	1.000

Fig. 5 The correlation analysis matrix for *RQ1*

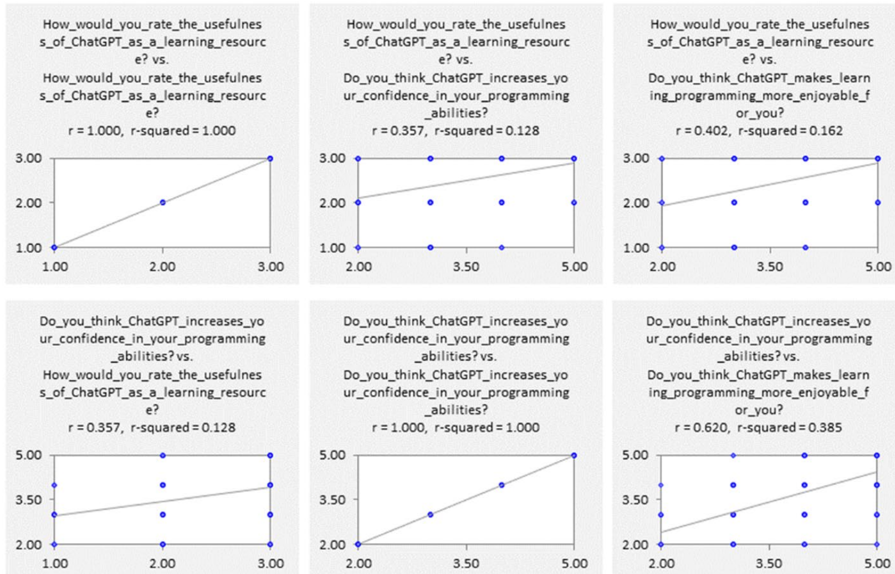


Fig. 6 The correlation analysis graph for *RQ1*

individuals who perceive ChatGPT as a valuable resource are more motivated to engage with programming concepts. However, the correlation between usefulness and *learning new programming concepts not covered in class* is weaker (r = 0.210r = 0.210). This suggests that while ChatGPT is useful, its impact on exploring unconventional or advanced topics is less pronounced.

Interestingly, the correlation between usefulness and *prior experience with programming* (r = 0.090r = 0.090) is very weak, indicating that the perceived value of ChatGPT as a learning resource is largely independent of participants' prior exposure to programming. Similarly, the negligible correlation with *prior usage of ChatGPT* (r = 0.092r = 0.092) suggests that previous familiarity with the tool has little influence on its perceived usefulness. This finding reinforces the notion that ChatGPT's usefulness is accessible to both novice and experienced users alike.

Other variables, such as *confidence in programming abilities* and *enjoyment of learning*, exhibit strong interconnections, influencing their relationship with the dependent variable. For instance, confidence strongly correlates with both enjoyment (r = 0.620r = 0.620) and motivation (r = 0.657r = 0.657), which in turn shape how participants perceive ChatGPT's usefulness. This interdependency highlights a broader ecosystem of positive learning outcomes facilitated by ChatGPT.

Therefore, the analysis indicates that the *usefulness of ChatGPT as a learning resource* is closely linked to confidence, enjoyment, and motivation among learners. However, its role in facilitating learning new programming concepts outside the classroom or relying on prior experience is limited. These findings suggest that while ChatGPT can significantly enhance conventional learning experiences, its impact on advanced or novel concept acquisition may require additional support or complementary resources.

4.1.3 Regression analysis on RQ1

Regression analysis (see Fig. 7) for the usefulness of ChatGPT as a learning resource provides insights into the factors influencing the dependent variable, *How would you rate* *the usefulness of ChatGPT as a learning resource?* This model evaluates the relationships between the dependent variable and six predictors, shedding light on what drives users' perceptions of ChatGPT's usefulness in learning.

The model explains 19.8% of the variance in the dependent variable, as indicated by the R-squared value of 0.198, as shown in Fig. 7. This suggests a low-to-moderate fit, implying that other factors outside the scope of this model may also influence perceptions of ChatGPT's usefulness. The adjusted R-squared value of 0.193 confirms that the inclusion of predictors does not artificially inflate the model's explanatory power. The standard error of regression (0.523) indicates the average deviation of predicted values from actual observations, reflecting moderate prediction accuracy.

Among the predictors, several variables emerged as significant contributors to the perceived usefulness of ChatGPT. The strongest predictor is whether respondents think ChatGPT makes learning programming more enjoyable. This variable has the largest coefficient (0.203) and the highest standardized coefficient (0.254), demonstrating a substantial positive relationship. It is followed by prior use of ChatGPT for any purpose, which also exerts a significant positive influence (coefficient = 0.351), suggesting that familiarity with the tool enhances perceptions of its utility. Confidence in programming abilities (coefficient = 0.086), motivation to learn programming (coefficient = 0.077), and learning new concepts not covered in class (coefficient = 0.090) also contribute positively and significantly to perceptions of usefulness. Each of these variables shows moderate but meaningful effects, supported by statistically significant t-statistics and low *p*-values.

One predictor, prior experience with programming, did not show a significant relationship with the usefulness rating. Its coefficient (– 0.022) and *p*-value (0.339) suggest that the level of programming experience before using ChatGPT does not meaningfully impact how useful it is perceived. This finding aligns with the idea that ChatGPT may be perceived as equally useful across varying levels of prior experience.

The variance inflation factor (VIF) values for all predictors are below the critical threshold of 5, confirming the absence of severe multicollinearity. The highest VIF is 2.153 for confidence in programming abilities, indicating a low level of correlation among predictors. This supports the robustness of the model and ensures that each predictor contributes independently to explaining the dependent variable.

An evaluation of error distribution reveals that the residuals are not normally distributed, as shown by the Anderson–Darling test (A-D stat = 11.46, p = 0.000). This deviation from normality could affect the reliability of *p*-values, but it does not invalidate the overall findings. The mean error of 0.000 indicates an unbiased model, while the root mean squared error (RMSE) of 0.521 reflects moderate residual variance.

Overall, the analysis highlights the importance of factors such as enjoyment, prior use, confidence, motivation, and exposure to new concepts in shaping the perceived

Regression Statistics: Model 2 for How_would_you_rate_the_usefulness_of_ChatGPT_as_a_learning_resource?(6 variables, n=855)

R-Squared	Adj.R-Sqr.	Std.Err.Reg.	Std.Dep.Var.	# Fitted	# Missing	Critical t	Confidence
0.198	0.193	0.523	0.582	855	444	1.963	95.0%

Coefficient Estimates: Model 2 for How_would_you_rate_the_usefulness_of_ChatGPT_as_a_learning_resource?(6 variables, n=855)

Variable	Coefficient	Std.Err.	t-Statistic	P-value	Lower95%	Upper95%	VIF	Std. Coeff.
Constant	0.732	0.180	4.057	0.000	0.378	1.086	0.000	0.000
Do_you_think_ChatGF	0.086	0.033	2.579	0.010	0.020	0.151	2.153	0.116
Do_you_think_ChatGF	0.203	0.033	6.207	0.000	0.139	0.267	1.766	0.254
Do_you_think_using_(	0.077	0.031	2.497	0.013	0.016	0.137	1.878	0.105
Does_ChatGPT_help_\	0.090	0.041	2.232	0.026	0.011	0.170	1.144	0.073
Have_you_used_Chat	0.351	0.141	2.488	0.013	0.074	0.628	1.005	0.077
How_would_you_rate	-0.022	0.023	-0.956	0.339				

Fig. 7 Regression analysis for the usefulness of ChatGPT as a learning resource

usefulness of ChatGPT as a learning resource. While the model captures some key relationships, the relatively low R-squared value suggests that additional factors not included in the model may also influence perceptions. Further exploration could involve identifying and analyzing these additional variables to enhance the model's explanatory power.

4.2 Evaluating ChatGPT's impact on student programming skills and learning (*RQ2*)

This section presents the findings regarding the impact of ChatGPT on students' programming skills and learning outcomes, specifically evaluating its influence on assignment completion, conceptual understanding, problem-solving capabilities, and overall course success. The analysis incorporated descriptive statistics, correlation analysis, and regression analysis to reveal the effects of ChatGPT on student programming skills.

4.2.1 Descriptive analysis on RQ2

The dataset comprises 855 observations, focusing on students' perceptions of ChatGPT's role in improving programming skills and engagement with course material. The descriptive statistics reveal important trends across these variables.

The average student score is 7.213, with a median of 9.000, suggesting that the majority of scores are relatively high, though the variability, indicated by a standard deviation of 3.566, points to some differences in student performance. Scores range from 0.000 to 10.000, reflecting a wide spectrum of achievement.

When asked whether ChatGPT improved problem-solving skills in programming, the average response was 3.374, with a median of 3.000, indicating a moderate level of agreement. The standard deviation of 0.854 shows some variability in responses, with answers ranging from 1.000 (Strongly Disagree) to 5.000 (Strongly Agree).

Students were also asked if ChatGPT affected their engagement with the course material. The mean response was 2.729, with a median of 3.000, suggesting mixed opinions. While some students agreed ChatGPT had an impact, others appeared neutral or disagreed, as reflected in the standard deviation of 0.953 and the range from 1.000 to 5.000.

Regarding whether ChatGPT helps in completing programming assignments, the mean score was 3.015, with a median of 3.000. This indicates a slightly positive perception overall, though the standard deviation of 1.042 suggests varied reliance among students, with responses spanning the full range of 1.000–5.000.

Students rated ChatGPT's effectiveness in helping them understand programming concepts with an average score of 3.409 and a median of 3.000. This shows a generally positive outlook, with less variability (standard deviation 0.776) compared to other variables. Responses to whether ChatGPT aids in understanding and fixing code errors followed a similar pattern, with a mean score of 3.407 and a standard deviation of 0.883, again indicating overall agreement but with some variation among students.

The availability of ChatGPT was deemed important to students' success in the course, as evidenced by a mean score of 3.662 and a median of 4.000. This suggests that many students see ChatGPT as a valuable resource. The relatively low standard deviation of 0.822 indicates consistency in this perception, with responses falling between 1.000 and 5.000.

Finally, the frequency of reliance on ChatGPT for assignments had an average response of 2.987 and a median of 3.000, showing moderate reliance overall. The variability, with a standard deviation of 0.921, reflects differing levels of dependence among students.

Descriptive Statistics

Variable	Stats 2 # Fitted	Mean	Median	Std.Dev.	Root.M.Sqr.	Std.Err.Mean	Minimum	Maximum	Skewness	Kurtosis
Overall__how_would_you_rate_the_impact_of_ChatGPT_on_students__programming_skills?	17	3.412	3.000	0.939	3.531	0.228	2.000	5.000	0.032	-0.670
Do_students_demonstrate_a_deeper_understanding_of_problem_solving_concepts_when_they_use_ChatGPT?	17	2.529	2.000	0.800	2.646	0.194	1.000	4.000	0.308	-0.152
Do_you_believe_students_who_use_ChatGPT_demonstrate_a_strong_understanding_of_programming_concepts_in_their_assignments?	17	2.588	3.000	1.004	2.765	0.243	1.000	4.000	-0.273	-0.813
Do_you_find_that_students_rely_on_ChatGPT_generated_solutions_without_fully_understanding_them?	17	4.294	4.000	0.849	4.372	0.206	2.000	5.000	-1.344	2.050
Have_you_noticed_any_changes_in_students__approach_to_problem_solving_since_the_introduction_of_ChatGPT?	17	3.765	4.000	1.147	3.926	0.278	1.000	5.000	-0.887	0.609
How_often_do_you_observe_significant_similarities_in_the_codes_submitted_by_different_students?	17	3.941	4.000	0.748	4.007	0.181	3.000	5.000	0.099	-1.047
In_your_opinion__has_the_use_of_ChatGPT_affected_students__ability_to_debug_and_troubleshoot_their_own_code?	17	3.353	3.000	1.272	3.573	0.308	2.000	5.000	0.270	-1.663

Fig. 8 The correlation analysis matrix for *RQ2*

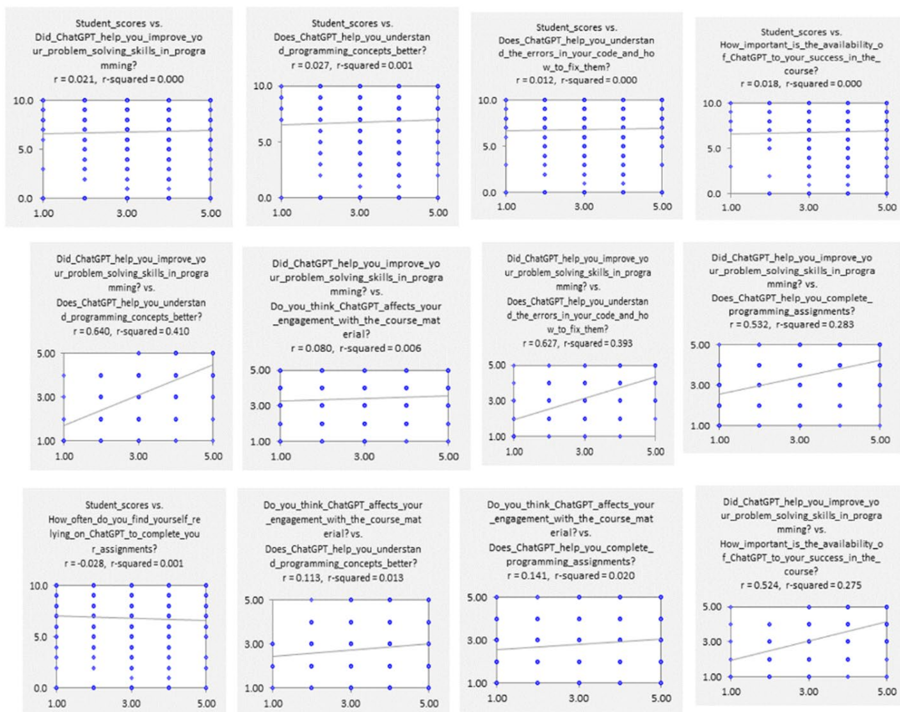


Fig. 9 The correlation analysis graph for *RQ2*

These descriptive statistics provide a foundational understanding of how students perceive ChatGPT's role in their programming education. Further analysis, such as examining the correlation matrix, could offer deeper insights into relationships between variables, such as whether reliance on ChatGPT or its perceived effectiveness significantly impacts student scores.

4.2.2 Correlation analysis on RQ2

The correlation matrix and graph in Figs. 8 and 9 provide insights into the relationships between students' scores and various factors related to their use of ChatGPT in programming tasks.

The results indicate a weak correlation between student scores and the question, *"Did ChatGPT help you improve your problem-solving skills in programming?"* (r = 0.021), suggesting minimal direct association. Similarly, engagement with course material shows a slight negative correlation with student scores (r = – 0.043), indicating that perceived impact on engagement does not significantly translate to improved academic performance.

Among the variables, the strongest relationships are observed between questions related to ChatGPT's specific functionalities. For instance, a high positive correlation exists between *"Does ChatGPT help you understand programming concepts better?"* and

"Did ChatGPT help you improve your problem-solving skills in programming?" (r = 0.640), as well as between *"Does ChatGPT help you understand the errors in your code and how to fix them?"* and programming concepts understanding (r = 0.577). These findings suggest a perceived interdependence of ChatGPT’s role in facilitating deeper conceptual understanding and problem-solving skills.

Interestingly, reliance on ChatGPT for completing assignments exhibits moderate correlations with factors such as assignment completion (r = 0.572) and the perceived importance of ChatGPT's availability (r = 0.412). This indicates that frequent reliance on ChatGPT may align with its perceived role in aiding coursework success, although this reliance does not strongly correlate with students' actual performance scores.

Overall, the matrix highlights that while students perceive ChatGPT as beneficial for specific programming-related tasks, these perceptions do not translate into significant, measurable improvements in academic performance as reflected in student scores.

The relatively low and, in some cases, negative adjusted R^2 values indicate that the models explain little of the variation in the dependent variables. This suggests that the predictors included may not strongly influence the outcomes or that other unmeasured factors play a larger role. While the models provide initial insights, their limited explanatory power highlights the need for further refinement and additional variables in future studies.

4.2.3 Regression analysis on RQ2

The regression analysis provides insights into the relationship between student scores (the dependent variable) and various independent variables that measure ChatGPT's impact on learning and academic performance. However, the model's performance suggests that these factors have minimal influence on student scores. The R-squared value of 0.006 indicates that the independent variables explain only 0.6% of the variance in student scores. This low value suggests that other factors not included in the model may be more significant in determining student performance.

None of the variables in the model were statistically significant at the 5% level, as indicated by their *p*-values, all of which were above 0.1. For instance, the variable "Did ChatGPT help you improve your problem-solving skills in programming?" has a positive coefficient of 0.100, suggesting a weak positive relationship with student scores. However, the relationship is not statistically significant, as the *p*-value is 0.504. Similarly, the variable "Do you think ChatGPT affects your engagement with the course material?" has a coefficient of − 0.146, indicating a slight negative association with scores, though this too is not significant (*p*-value = 0.103). Other variables, such as whether ChatGPT helps with understanding programming concepts or completing assignments, also show weak associations with student scores but lack statistical significance.

The model's Root Mean Squared Error (RMSE) of 3.457 highlights the extent to which observed student scores deviate from predicted values, suggesting a substantial amount of unexplained variability. Furthermore, the Adjusted R-squared value of 0.002 confirms that the inclusion of these predictors does not substantially improve the model's explanatory power. The Anderson–Darling test for residuals (A-D stat = 103.65, P = 0.000) indicates that the residuals are not normally distributed, which could affect the reliability of the estimated coefficients under standard regression assumptions.

In general, the analysis reveals that the selected variables related to ChatGPT's perceived benefits are not significant predictors of student scores. The low explanatory power of the model suggests that other unexamined factors, such as students' prior programming knowledge, study habits, or alternative learning tools, may be more influential. Future research could incorporate additional predictors, explore non-linear relationships, or refine the survey design to capture a broader range of influences on student performance.

4.3 Markers perceived ChatGPT's impact on student programming skills (*RQ3*)

This section presents the findings regarding markers' perceptions of ChatGPT's effect on student programming skills, specifically concerning conceptual understanding, debugging competence, and problem-solving approaches. The analysis includes descriptive statistics, correlation analysis, and regression analysis focused on this perceived impact.

4.3.1 Descriptive analysis on RQ3

The descriptive statistics provide insights into perceptions regarding the impact of ChatGPT on students' programming skills, understanding of problem-solving concepts, reliance on generated solutions, and their ability to debug and troubleshoot their own code. The responses were analyzed from 17 participants, with metrics such as mean, median, standard deviation, skewness, and kurtosis used to interpret the data, as shown in Figs. 10 and 11. Figure 10 presents the descriptive analysis matrix for *RQ3*, and Fig. 11 presents the descriptive analysis graph for RQ3.

The overall rating for the impact of ChatGPT on students' programming skills has a mean of 3.41, indicating a moderately positive perception. The responses are relatively consistent, as suggested by a standard deviation of 0.94 and minimal skewness (0.032). However, the slightly negative kurtosis (– 0.670) suggests that the responses are slightly flatter than a normal distribution.

When assessing whether students demonstrate a deeper understanding of problem-solving concepts when using ChatGPT, the mean is lower at 2.53, with a standard deviation of 0.80, indicating a moderate to low agreement. The positive skewness (0.308) suggests a slight bias toward lower ratings, while the kurtosis (– 0.152) points to a distribution that is close to normal.

Descriptive Statistics

Variable	Skewness	Kurtosis
Overall__how_would_you_rate_the_impact_of_ChatGPT_on_students__programming_skills?	0.032	-0.670
Do_students_demonstrate_a_deeper_understanding_of_problem_solving_concepts_when_they_use_ChatGPT?	0.308	-0.152
Do_you_believe_students_who_use_ChatGPT_demonstrate_a_strong_understanding_of_programming_concepts_in_their_assignments?	-0.273	-0.813
Do_you_find_that_students_rely_on_ChatGPT_generated_solutions_without_fully_understanding_them?	-1.344	2.050
Have_you_noticed_any_changes_in_students__approach_to_problem_solving_since_the_introduction_of_ChatGPT?	-0.887	0.609
How_often_do_you_observe_significant_similarities_in_the_codes_submitted_by_different_students?	0.099	-1.047
In_your_opinion__has_the_use_of_ChatGPT_affected_students__ability_to_debug_and_troubleshoot_their_own_code?	0.270	-1.663

# Fitted	Mean	Median	Std.Dev.	Root.M.Sqr.	Std.Err.Mean	Minimum	Maximum
17	3.412	3.000	0.939	3.531	0.228	2.000	5.000
17	2.529	2.000	0.800	2.646	0.194	1.000	4.000
17	2.588	3.000	1.004	2.765	0.243	1.000	4.000
17	4.294	4.000	0.849	4.372	0.206	2.000	5.000
17	3.765	4.000	1.147	3.926	0.278	1.000	5.000
17	3.941	4.000	0.748	4.007	0.181	3.000	5.000
17	3.353	3.000	1.272	3.573	0.308	2.000	5.000

Fig. 10 Descriptive analysis matrix for *RQ3*

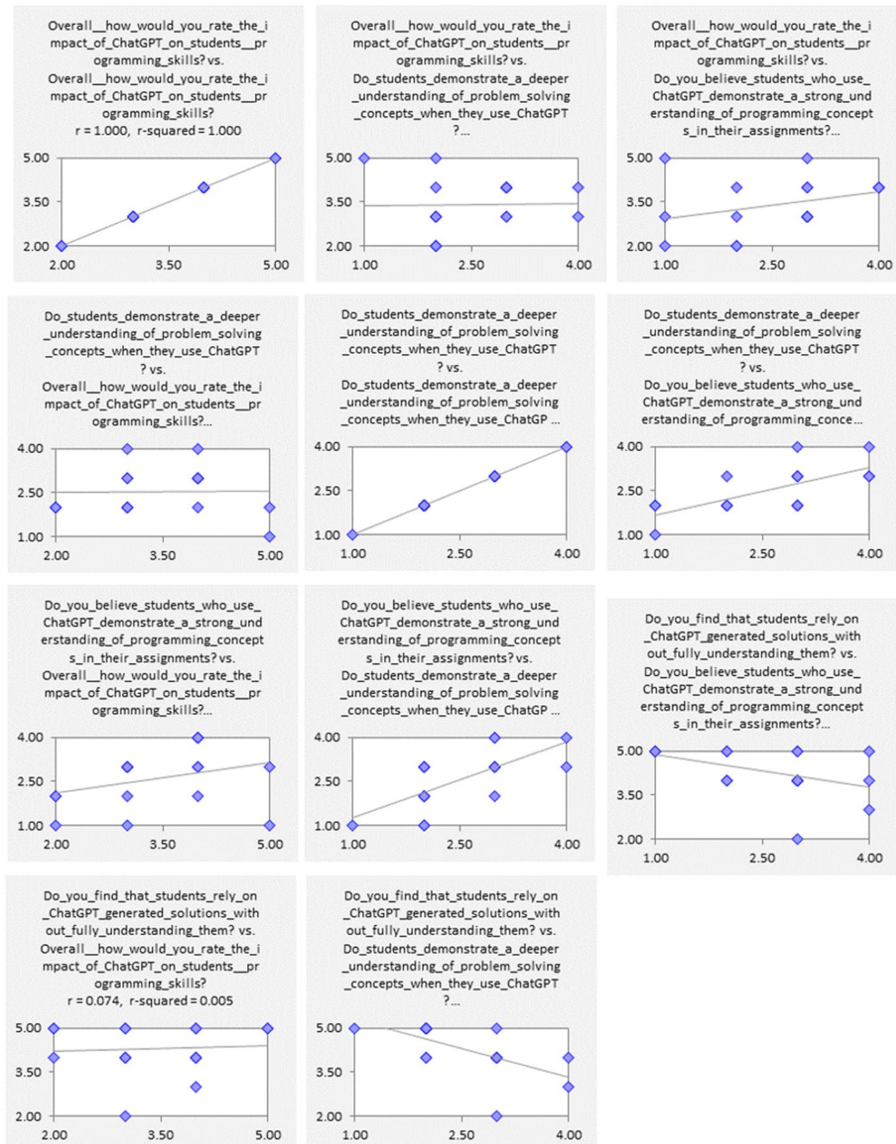


Fig. 11 Descriptive analysis graph for *RQ3*

Regarding students' understanding of programming concepts in their assignments, the mean is 2.59, close to the median of 3.00. This suggests mixed opinions. The standard deviation of 1.00 reflects moderate variability in responses, while the skewness (– 0.273) indicates a slight bias toward higher ratings. The kurtosis (– 0.813) implies a flatter-than-normal distribution.

There is strong agreement on students relying on ChatGPT-generated solutions without fully understanding them, as evidenced by a high mean of 4.29. The standard deviation of 0.85 and negative skewness (– 1.344) highlight a strong consensus on this issue, with a bias toward higher ratings. The kurtosis (2.050) further emphasizes the peaked nature of the responses, indicating that most respondents rated this concern highly.

Perceptions of changes in students' approach to problem-solving since the introduction of ChatGPT yielded a mean of 3.77, reflecting moderately high agreement. The standard deviation of 1.15 suggests some variability in opinions, while the skewness

(– 0.887) and kurtosis (0.609) indicate a slight bias toward higher ratings with a sharper distribution.

The observation of significant similarities in students' submitted codes has a high mean of 3.94 and a low standard deviation of 0.75, showing strong agreement among participants. The skewness (0.099) and kurtosis (– 1.047) suggest a relatively normal but slightly flatter distribution.

Lastly, opinions on whether ChatGPT has affected students' ability to debug and troubleshoot their own code yield a mean of 3.35, indicating moderate agreement. The higher standard deviation of 1.27 reflects more variability in responses, and the positive skewness (0.270) suggests a slight bias toward lower ratings. The kurtosis (– 1.663) indicates a flatter-than-normal distribution.

Overall, the results reveal nuanced perceptions of ChatGPT's impact, with strong concerns about reliance on AI-generated solutions and noticeable changes in students' problem-solving approaches. While some areas, like programming skills and debugging abilities, show moderate agreement, the variability in responses underscores the complexity of the topic.

4.3.2 Correlation analysis on RQ3

The correlation matrix in Fig. 12 offers *valuable insights* into the relationships between different perceptions of ChatGPT's impact on students' programming skills. First, the *overall rating of ChatGPT's impact on programming skills* exhibits weak positive correlations with most other variables. Notably, it shows a moderate positive correlation with *"Do you believe students who use ChatGPT demonstrate a strong understanding of programming concepts in their assignments?"* (0.324). This suggests that those who perceive ChatGPT as having a positive impact on programming skills also tend to believe it enhances students' understanding of programming concepts. Additionally, there is a weak positive correlation with *"Have you noticed any changes in students' approach to problem-solving since the introduction of ChatGPT?"* (0.328), indicating that respondents who rate ChatGPT positively also perceive *some changes* in students' problem-solving approaches.

The variable *"Do students demonstrate a deeper understanding of problem-solving concepts when they use ChatGPT?"* exhibits a moderately high correlation with *"Do you believe students who use ChatGPT demonstrate a strong understanding of programming concepts in their assignments?"* (0.678). This suggests that those who think ChatGPT aids in problem-solving are also likely to believe it improves students' understanding of programming concepts. Furthermore, it shows a strong negative correlation with *"Do you find that students rely on ChatGPT-generated solutions without fully understanding them?"* (– 0.612), indicating that respondents who perceive a deeper understanding of problem-solving concepts are less likely to believe students overly rely on ChatGPT solutions without comprehension.

Correlation Matrix (n=17)

Variable	Overall__how_would_you_rate_the_impact_of_ChatGPT_on_students__programming_skills?	Do_students_demonstrate_a_deeper_understanding_of_problem_solving_concepts_when_they_use_ChatGPT?	Do_you_believe_students_who_use_ChatGPT_demonstrate_a_strong_understanding_of_programming_concepts_in_their_assignments?	Do_you_find_that_students_rely_on_ChatGPT_generated_solutions_without_fully_understanding_them?	Have_you_noticed_any_changes_in_students__approach_to_problem_solving_since_the_introduction_of_ChatGPT?	How_often_do_you_observe_significant_similarities_in_the_codes_submitted_by_different_students?	In_your_opinion__has_the_use_of_ChatGPT_affected_students__ability_to_debug_and_tr
Overall__how_would_you_rate_the_impact_of_ChatGPT_on_students__programming_skills?	1.000						
Do_students_demonstrate_a_deeper_understanding_of_problem_solving_concepts_when_they_use_ChatGPT?	0.024	1.000					
Do_you_believe_students_who_use_ChatGPT_demonstrate_a_strong_understanding_of_programming_concepts_in_their_assignments?	0.324	0.678	1.000				
Do_you_find_that_students_rely_on_ChatGPT_generated_solutions_without_fully_understanding_them?	0.074	-0.612	-0.436	1.000			
Have_you_noticed_any_changes_in_students__approach_to_problem_solving_since_the_introduction_of_ChatGPT?	0.328	0.144	0.309	-0.117	1.000		
How_often_do_you_observe_significant_similarities_in_the_codes_submitted_by_different_students?	-0.319	-0.467	-0.534	0.324	-0.090	1.000	
In_your_opinion__has_the_use_of_ChatGPT_affected_students__ability_to_debug_and_troubleshoot_their_own_code?	0.185	0.112	0.317	-0.276	0.446	-0.043	1.000

Fig. 12 Correlation analysis matrix for *RQ3*

For the variable *"Do you find that students rely on ChatGPT-generated solutions without fully understanding them?"*, a weak negative correlation is observed with *"Do you believe students who use ChatGPT demonstrate a strong understanding of programming concepts in their assignments?"* (– 0.436). This highlights that *higher reliance* on ChatGPT-generated solutions without understanding is associated with a *lower perception* of students' understanding of programming concepts. Additionally, the negative correlation with *"Have you noticed any changes in students' approach to problem-solving since the introduction of ChatGPT?"* (– 0.117) implies that reliance on ChatGPT has minimal impact on perceived changes in students' problem-solving approaches.

The variable *"Have you noticed any changes in students' approach to problem-solving since the introduction of ChatGPT?"* correlates moderately with *"Do you believe students who use ChatGPT demonstrate a strong understanding of programming concepts in their assignments?"* (0.399). This indicates that respondents who observe *changes in problem-solving strategies* also tend to think ChatGPT enhances students' programming understanding. Negative correlations with *"How often do you observe significant similarities in the codes submitted by different students?"* (– 0.090) and *"Do you find that students rely on ChatGPT-generated solutions without fully understanding them?"* (– 0.117) suggest that observed changes in problem-solving strategies are not strongly linked to *code similarities* or *reliance on ChatGPT solutions.*

When examining *"How often do you observe significant similarities in the codes submitted by different students?"*, negative correlations emerge with *"Overall, how would you rate the impact of ChatGPT on students' programming skills?"* (– 0.319), *"Do you believe students who use ChatGPT demonstrate a strong understanding of programming concepts in their assignments?"* (– 0.534), and *"Do you find that students rely on ChatGPT-generated solutions without fully understanding them?"* (0.324). These findings indicate that respondents who frequently observe high *similarities in code submissions* tend to rate ChatGPT's impact less positively and believe students struggle to fully grasp programming concepts when using ChatGPT.

Finally, *"In your opinion, has the use of ChatGPT affected students' ability to debug and troubleshoot their own code?"* shows weak positive correlations with *"Overall, how would you rate the impact of ChatGPT on students' programming skills?"* (0.185) and *"Do you believe students who use ChatGPT demonstrate a strong understanding of programming concepts in their assignments?"* (0.317). These correlations suggest that those who believe ChatGPT aids debugging and troubleshooting tend to rate its overall impact and programming benefits more positively. A weak negative correlation with *"Do you find that students rely on ChatGPT-generated solutions without fully understanding them?"* (– 0.276) further indicates that respondents who see improvements in debugging skills believe students are *less reliant* on ChatGPT solutions.

In summary, the correlation matrix reveals valuable relationships between perceptions of ChatGPT's role in programming development. The most significant findings highlight that reliance on ChatGPT without full understanding negatively correlates with perceived benefits in programming and problem-solving abilities. Conversely, respondents who observe changes in problem-solving approaches or improvements in debugging skills tend to rate ChatGPT's overall impact more favourably.

	R-Squared	Adj.R-Sqr.	Std.Err.Reg.	Std.Dep.Var.	# Fitted	# Missing	Critical t	Confidence
	0.288	-0.139	1.003	0.939	17	0	2.228	95.0%

Adj. R-sqr. is negative because the standard error of the regression is greater than the standard deviation of the dependent variable.

Variable	Coefficient	Std.Err.	t-Statistic	P-value	Lower95%	Upper95%	VIF	Std. Coeff.
Constant	3.184	3.232	0.985	0.348	-4.017	10.385	0.000	0.000
Do_students_demonst	-0.285	0.506	-0.563	0.586	-1.412	0.843	2.606	-0.242
Do_you_believe_stude	0.286	0.409	0.699	0.500	-0.625	1.196	2.676	0.305
Do_you_find_that_stu	0.224	0.390	0.575	0.578	-0.645	1.093	1.745	0.203
Have_you_noticed_an	0.169	0.261	0.650	0.530	-0.411	0.750	1.423	0.207
How_often_do_you_c	-0.394	0.409	-0.964	0.358	-1.306	0.517	1.489	-0.314
In_your_opinion__has	0.048	0.235	0.205	0.841	-0.475	0.572	1.420	0.065

	Mean Error	RMSE	MAE	Minimum	Maximum	MAPE	A-D* stat
Fitted (n=17)	0.000	0.769	0.584	-1.723	1.577	20.3%	0.41 (P=0.345)

Fig. 13 Regression analysis results of *RQ3*

4.3.3 Regression analysis on RQ3

The regression analysis results in Fig. 13 provide insights into how various factors influence the perceived impact of ChatGPT on students' programming skills by Lecturers. The model's dependent variable is *"Overall, how would you rate the impact of ChatGPT on students' programming skills?"*, with the independent variables encompassing different aspects such as students' understanding of problem-solving concepts, reliance on ChatGPT-generated solutions, changes in students' approach to problem-solving, and the effect on their ability to debug and troubleshoot code.

The model's R-squared value of 0.288 indicates that approximately 28.8% of the variability in the perceived impact of ChatGPT on programming skills is explained by the independent variables included in the model. However, the Adjusted R-squared value is negative (− 0.139), which suggests that the model may not be a good fit for the data, as the standard error of the regression is larger than the standard deviation of the dependent variable. The Standard Error of the Regression is 1.003, and the Standard Deviation of the Dependent Variable is 0.939, indicating that the model's predictions are less precise than the inherent variability in the data.

The regression analysis provides insights into how various factors influence the perceived impact of ChatGPT on students' programming skills. The dependent variable in the model is *"Overall, how would you rate the impact of ChatGPT on students' programming skills?"*, while the independent variables include factors such as students' understanding of problem-solving concepts, reliance on ChatGPT-generated solutions, changes in problem-solving approaches, and the effect on their debugging and troubleshooting abilities.

The model's R-squared value of 0.288 indicates that the independent variables explain approximately 28.8% of the variability in the perceived impact of ChatGPT on programming skills. However, the negative Adjusted R-squared value of − 0.139 suggests the model may not be a good fit, as the regression's standard error exceeds the dependent variable's standard deviation. Specifically, the Standard Error of the Regression is 1.003, while the Standard Deviation of the Dependent Variable is 0.939, highlighting less precision in the model's predictions compared to the inherent variability in the data.

The coefficient for the intercept is 3.184, representing the baseline perception of ChatGPT's impact when all independent variables are zero. However, the intercept is not statistically significant, with a t-statistic of 0.985 and a *p*-value of 0.348. Among the independent variables, *"Do students demonstrate a deeper understanding of problem-solving concepts when they use ChatGPT?"* has a coefficient of − 0.285, suggesting a weak negative relationship with the perceived impact. Yet, this relationship is not statistically significant with a t-statistic of – 0.563 and a p-value of 0.586. Similarly, *"Do you believe students who use ChatGPT demonstrate a strong understanding of programming concepts in their assignments?"* shows a weak positive association with a coefficient of 0.286, but this, too, lacks statistical significance (t-statistic = 0.699, p-value = 0.500).

The variable *"Do you find that students rely on ChatGPT-generated solutions without fully understanding them?"* has a coefficient of 0.224, indicating a slight positive relationship, but its t-statistic of 0.575 and p-value of 0.578 suggest that reliance on ChatGPT does not significantly influence perceptions of its impact. Similarly, *"Have you noticed any changes in students' approach to problem-solving since the introduction of ChatGPT?"* has a small positive coefficient of 0.169 but is not statistically significant (t-statistic = 0.650, p-value = 0.530). In contrast, the variable *"How often do you observe significant similarities in the codes submitted by different students?"* has a negative coefficient of – 0.394, suggesting that greater similarities in student submissions are associated with less positive perceptions of ChatGPT's impact. However, this relationship is also not statistically significant (t-statistic = – 0.964, p-value = 0.358). Lastly, the coefficient for *"In your opinion, has the use of ChatGPT affected students' ability to debug and troubleshoot their own code?"* is 0.048, indicating a very weak positive relationship. This effect is likewise insignificant, with a t-statistic of 0.205 and a p-value of 0.841.

The model's overall prediction accuracy is moderate. The Mean Error of 0.000 suggests no systematic bias in the model's predictions. However, the Root Mean Square Error (RMSE) of 0.769 and the Mean Absolute Error (MAE) of 0.584 indicate notable prediction errors. The Minimum and Maximum fitted values range from – 1.723 to 1.577, while the Mean Absolute Percentage Error (MAPE) of 20.3% reveals that, on average, the predictions deviate from actual values by 20.3%.

In conclusion, the regression model's findings suggest that none of the independent variables significantly predict the perceived impact of ChatGPT on students' programming skills. The low significance levels across all coefficients imply that the relationships between the independent variables and the dependent variable are weak. Furthermore, the negative Adjusted R-squared value suggests that the model does not adequately capture the variability in the data, indicating potential limitations in the model's predictive power. Future research could improve the model by incorporating additional variables, refining survey questions, or exploring non-linear relationships to better capture the factors influencing the perceived impact of ChatGPT.

5 Strengths and limitations

The paper demonstrates several strengths in its investigation of ChatGPT's impact on programming education. Notably, it employs a mixed-methods research approach, combining quantitative analysis of student performance metrics with qualitative assessment of learning experiences. This provides a more comprehensive understanding of the phenomenon than relying solely on one type of data. The study also benefits from a large sample size of 1889 undergraduate students from diverse faculties within a mandatory Python programming course, enhancing the breadth of perspectives captured. Furthermore, the collection of data from both students through surveys and the course instructor offers a balanced view of the impact. The survey instrument was carefully crafted with both closed-ended and open-ended questions to gather in-depth insights into various aspects of ChatGPT usage, including frequency, perceptions, effectiveness, and challenges. Finally, the paper outlines a rigorous statistical analysis methodology, including checks for adherence to Ordinary Least Squares (OLS) regression assumptions, aiming to ensure the reliability and validity of the quantitative findings.

Despite these strengths and valuable contributions, this paper acknowledges several limitations that should be considered when interpreting the findings. First, the research employed a cross-sectional survey design, which restricts the ability to establish causal relationships between the use of ChatGPT and learning outcomes. Since the data were collected at a single point in time, the long-term effects of ChatGPT on students' programming skills and academic performance remain uncertain. Furthermore, the study relied primarily on self-reported data from students, which is inherently prone to biases such as social desirability and subjective interpretation, potentially affecting the accuracy of reported perceptions and experiences.

Another important limitation concerns the research context. The study was conducted within a specific institutional and cultural setting—a Python programming course at a single university in Nigeria. This narrow focus may constrain the generalisability of the findings to other programming languages, educational contexts, or cultural environments. In addition, the relatively small sample size of instructors limits the robustness of conclusions drawn regarding educator perspectives and instructional experiences.

From a statistical standpoint, several analytical constraints were observed. The low R-squared values in some regression models suggest that the variables included in the analysis explain only a small portion of the variance in student performance and instructor perceptions, indicating that other unmeasured factors may significantly influence these outcomes. Moreover, the presence of non-normal residuals in certain analyses points to potential violations of statistical assumptions, which could affect the precision of the reported p-values. The study also encountered ceiling effects in the performance variable, where high baseline scores may have limited the detection of further improvement.

Overall, while the findings provide meaningful insights into the educational application of ChatGPT, future studies could adopt experimental or longitudinal designs to validate these results more rigorously. Expanding the research across multiple institutions, disciplines, and cultural contexts would also enhance the external validity and provide a more comprehensive understanding of ChatGPT's role in promoting learning outcomes.

6 Recommendations for future work

Building on the limitations identified in this study, several avenues for future research are recommended to deepen the understanding of ChatGPT's role in programming education and enhance the robustness of empirical findings.

First, methodological improvements are essential. The correlational and cross-sectional nature of the present study limits causal inference; therefore, future investigations should adopt experimental or quasi-experimental designs to establish clearer causal relationships between specific modes of ChatGPT use and learning outcomes, such as conceptual understanding, debugging efficiency, and code quality. Longitudinal studies would also be valuable in examining the sustained influence of ChatGPT on students' skill development, critical thinking, and potential behavioral patterns, particularly the risk of over-reliance on AI tools [69].

Second, research should explore the cognitive processes underlying students' interactions with ChatGPT. Qualitative approaches, including think-aloud protocols, screen recordings, or interaction log analyses, can provide deeper insights into whether learners are actively engaging with AI support or merely delegating cognitive effort. Such investigations could also assess how ChatGPT affects higher-order skills such as critical thinking, reasoning, and complex problem-solving over time [72].

Third, future work should investigate pedagogical strategies for effectively integrating ChatGPT into programming education. Comparative studies could evaluate the effectiveness of using ChatGPT in structured activities, such as code debugging and concept clarification, versus open-ended exploratory tasks. Additionally, assignments that encourage students to critically evaluate, refine, or justify AI-generated content could foster deeper learning and more responsible engagement with generative AI tools [70].

Fourth, the importance of contextual diversity should be emphasized in subsequent research. Replicating this study across multiple institutions, programming languages, and course levels, from introductory to advanced, would provide a broader understanding of the generalisability of current findings. Cultural factors influencing the adoption and perception of ChatGPT should also be examined, as these may shape how learners and instructors interact with AI-assisted learning systems in different educational and socio-cultural settings [51, 71].

Finally, future studies should focus on assessment innovation and academic integrity within AI-supported learning environments. There is a need to develop novel assessment frameworks that can distinguish genuine student understanding and problem-solving skills from AI-assisted outputs. Equally important is the exploration of pedagogical and technological mechanisms to uphold academic integrity and prevent over-reliance on AI, ensuring that ChatGPT remains a tool for augmentation rather than substitution of learning.

Collectively, these directions would not only strengthen empirical evidence on ChatGPT's educational impact but also inform the development of pedagogical frameworks that balance innovation with integrity in the era of generative AI.

7 Conclusions

This study investigated the multifaceted impact of ChatGPT on student learning outcomes and performance within the context of introductory computer programming courses. Confronting the rapid integration of generative AI into education, the research addressed the critical need for empirical evidence regarding ChatGPT's effects, particularly in relation to student skill development, conceptual understanding, and academic integrity.

The findings reveal a nuanced picture. Students largely perceive ChatGPT positively, reporting increased enjoyment, confidence, and motivation in learning programming. They find it valuable for specific tasks, such as understanding concepts and debugging code, viewing its availability as crucial for their course success. However, a significant disconnect emerges between these positive affective perceptions and measurable academic performance, as evidenced by course scores. Statistical analyses revealed limited predictive power of ChatGPT usage or related perceptions on final grades ($R^2 = 0.006$; no predictors were significant).

Instructor observations raise pertinent concerns regarding potential student over-reliance on AI-generated solutions, similarities in submitted code, and potential negative impacts on independent problem-solving abilities and deeper conceptual grasp when used without sufficient cognitive engagement. This suggests that while ChatGPT enhances learning experiences in some ways, an excessive reliance on the tool may foster surface-level learning and hinder the development of critical thinking.

Consequently, this research highlights that while ChatGPT offers tangible benefits in terms of student engagement and perceived support, its integration into programming education does not automatically translate into enhanced learning outcomes or performance. The disconnect between perceived utility and demonstrable skill acquisition emphasizes the need for careful and structured integration of ChatGPT into educational contexts. As such, this study makes an important contribution by empirically demonstrating this complex interplay, providing valuable insights for educators who navigate the integration of AI tools.

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Author contributions
The study was conceptualized by I.G., O.G., and O.J.O., who also designed the core methodology. O.G. led the investigation and formal analysis, with significant contributions to the investigation and data curation from W.O.S., O.J.A., and R.E.S. O.J.A. managed project administration and resources, and R.E.S. performed validation. The initial draft of the manuscript and the visualizations were prepared by I.G. and O.G. R.M. contributed to the software implementation, visualization, and editing of the manuscript. Supervision for the project was provided by I.G. All authors participated in the review and editing process and have read and approved the final manuscript.

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Data availability
The data that support the findings of this study were collected from student participants under the assurance of confidentiality and anonymity. As such, the raw dataset is not publicly available to protect participant privacy. However, an anonymized version of the data can be made available by the corresponding author upon a reasonable request.

Declarations

Ethics approval and consent to participate
This study was conducted in accordance with the ethical standards for research involving human subjects. The research protocol received formal ethical clearance and approval from the Institutional Review Committee of Obafemi Awolowo University, Nigeria, prior to the commencement of data collection. The study adhered to established procedures for data handling and confidentiality, ensuring that all collected information was anonymized and utilized strictly for the academic purposes outlined in the approved research protocol.

Consent for publication
Not applicable. This manuscript does not contain any individual person's data in any identifiable form.

Informed consent
Informed consent was obtained from all individual participants (students and instructors) included in the study.
Prior to accessing the survey instrument, participants were presented with a digital briefing that detailed the study's objectives, the voluntary nature of their involvement, and their right to withdraw at any time without penalty. Consent was documented through the participants' voluntary progression to the survey questions following this introductory statement. No personally identifiable information was collected, ensuring that consent was provided on the basis of complete anonymity.

Competing interests
The authors declare no competing interests.

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